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5.4.3 Cholesky factorization algorithm (right-looking variant)

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from

that we can read off what the operations are that must happen in the

current step and in the future. Namely we know that α_{11} is equal to λ_{11} squared and we know what α_{11} is. And therefore we know that λ_{11}

must be plus or minus the square root of α_{11} . And we choose to make it equal

to a positive number. And therefore we take it to be equal to the square root of α_{11} . Now. wait

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Reading assignment

1/1 point (graded)



Read Unit 5.4.3 of the notes. [\[LINK\]](#)

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[Typo in Homework 5.4.3.1](#)

According to the lecture and the textbook, shouldn't the update of A22 be: ...

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Homework 5.4.3.1

1/1 point (graded)
Choose the approximate cost incurred by

$$A = \text{Chol-right-looking}(A)$$
$$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$$
$$A_{TL} \text{ is } 0 \times 0$$
$$\textbf{while } n(A_{TL}) < n(A)$$
$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$

$$\alpha_{11} := \lambda_{11} = \sqrt{\alpha_{11}}$$
$$a_{21} := l_{21} = a_{21}/\alpha_{11}$$
$$A_{22} := A_{22} - a_{21}a_{21}^H \quad (\text{syr: update only lower triangular part})$$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$
$$\textbf{endwhile}$$

when applied to an $n \times n$ matrix.

- ☐ n^3
- ☐ $\frac{2}{3}n^3$
- ☒ $\frac{1}{3}n^3$



Solution:

For a complete solution, see [the notes](#).

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Answers are displayed within the problem

Homework 5.4.3.2

1/1 point (graded)
Implement the algorithm

$$A = \text{Chol-right-looking}(A)$$
$$A \rightarrow \left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$$
$$A_{TL} \text{ is } 0 \times 0$$
$$\textbf{while } n(A_{TL}) < n(A)$$
$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$

$$\alpha_{11} := \lambda_{11} = \sqrt{\alpha_{11}}$$
$$a_{21} := l_{21} = a_{21}/\alpha_{11}$$
$$A_{22} := A_{22} - a_{21}a_{12}^T \quad (\text{syr: update only lower triangular part})$$

$$\left(\begin{array}{c|c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c|c|c} A_{00} & a_{01} & A_{02} \\ \hline a_{10}^T & \alpha_{11} & a_{12}^T \\ A_{20} & a_{21} & A_{22} \end{array} \right)$$
$$\textbf{endwhile}$$

as

function [A_out] = Chol_right_looking(A)

by completing the code in
[Assignments/Week05/matlab/Chol_right_looking.m](#). Input is a HPD
 $m \times n$ matrix A with only the lower triangular part stored. Output is
the matrix A that has its lower triangular part overwritten with the
Chol factor L .

Cholesky factor. You may want to use [Assignments/Week05/matlab/test_Chol_right_looking.m](#) to check your implementation.

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